FYP Project Proposal

Project BagSense: An Edge-IoT Multi-Tag Content Identification and Real-Time Inventory Monitoring Smart Bag with Dual Physical-Mobile Interface

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Abstract

Students and professionals frequently encounter daily disruptions due to forgotten, misplaced, or unorganized personal belongings—such as student identification badges, laptop chargers, laboratory kits, and medication. Existing commercial tracking accessories (e.g., Apple AirTags, Tile) require active, battery-powered hardware per item, imposing prohibitive scaling costs. This proposal introduces **Project BagSense**, an intelligent, energy-efficient Edge-IoT smart backpack system capable of autonomous, non-line-of-sight inventory detection inside the bag cavity. Operating via near-field Ultra-High Frequency (UHF) passive RFID interrogation paired with hall-effect zipper state sensing, the system inventories tagged personal belongings dynamically without requiring line-of-sight. The bag provides immediate visual feedback through both an **external low-power E-Paper display** mounted on the outer chassis and a companion **Flutter-based mobile dashboard** communicating over Bluetooth Low Energy (BLE) and Wi-Fi. Designed, simulated, and prototyped at an estimated cost of under RM 600, BagSense demonstrates exceptional practical utility, cross-disciplinary technical depth, and strong alignment with IIUM's Smart Campus research agenda.

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1 Project Background & Problem Statement

The daily routine of university students and academic staff across the KICT complex entails frequent transitions between lecture halls, research laboratories, study areas, and residential colleges. A pervasive problem is the unintentional misplacement or omission of critical everyday gear (e.g., matric cards, scientific calculators, portable hard drives, prescription glasses, and power adapters).

Current approaches to personal item tracking suffer from critical structural weaknesses:

1. **High Per-Item Cost & Battery Dependence:** Active Bluetooth beacon tags (e.g., AirTags, SmartTags) cost RM 120–RM 160 per unit and rely on internal coin cells. Tagging ten common items within a backpack demands over RM 1,200 solely in consumable tags.
2. **Optical Occlusion in Closed Bags:** Optical camera-based approaches inside bags fail due to extreme occlusions, lack of ambient illumination, and erratic item folding.
3. **Passive and Friction-Heavy Verification:** Traditional packing requires manual physical searching or opening the bag. Users often only discover a missing item when they have already arrived at their lecture or examination venue.

Proposed Solution: BagSense solves this by embedding an ultralow-power UHF RFID reader array within an electromagnetically shielded fabric cavity liner. Passive RFID tags (costing less than RM 0.50 each, paper-thin, battery-free, and virtually indestructible) are affixed to personal belongings. An embedded micro-controller triggers automated inventory scans when the bag's zipper closes, projecting the updated manifest onto an exterior glanceable E-Paper display and syncing seamlessly with a smartphone application.

2 Project Objectives

The core technical objective is to design, model, and deploy a low-cost, production-feasible smart backpack that automates item awareness. The specific measurable objectives are:

1. **Cavity Ergonomics & 3D Enclosure Modeling:** To model the internal compartment liner, Faraday-damping lining, and modular electronics casing in Blender/CAD to ensure aesthetic discretion, weatherproofing, and user comfort.
2. **Multi-Tag Non-Line-of-Sight Inventory Engine:** To implement and optimize anti-collision UHF RFID interrogation capable of scanning and identifying up to 15 tagged items within the enclosed bag cavity in under 1.5 seconds with $\geq 98\%$ recall.
3. **Dual-Interface Visualization Framework:** To develop a low-power external E-Paper display (glanceable status outside the bag) and a cross-platform mobile application (Flutter) for interactive checklist creation, missing item push notifications, and schedule-based packing prompts.
4. **Power-Aware Edge Firmware:** To design an event-driven ESP32-S3 firmware architecture utilizing deep-sleep states and magnetic reed/hall-effect zipper switches to achieve an operational endurance exceeding 5 days per charge.
5. **Field Validation in Campus Environment:** To benchmark recognition accuracy, signal leakage boundaries, and usability across real-world testing scenarios within the KICT building.

3 Project Work Breakdown (9-Month Roadmap)

The project follows a rigorous 3-phase engineering cycle structured across nine calendar months.

4 Software Architecture and Toolchain

The software ecosystem combines modern embedded systems programming, computer-aided design, and modern mobile app engineering:

- **Blender & Fusion 360:** Utilized for 3D modeling of the protective module housing, snap-fit display bezel for the outer bag pocket, and internal cable conduit sleeves.
- **VS Code with PlatformIO / ESP-IDF:** Primary development environment for firmware engineering, utilizing FreeRTOS tasks to decouple sensor polling, display refreshes, and BLE state notifications.
- **Flutter Framework (Dart):** Cross-platform mobile front-end providing a polished user interface, featuring interactive category filters, custom luggage packing lists, and visual alert banners.

Phase	Core Focus Area	Key Deliverables & Milestones
Phase 1: Pre-Dev (Months 1–3)	• RF Field & Cavity Attenuation Study • 3D CAD Modeling & Enclosure Design • Mobile UI/UX Wireframing	• Comprehensive FYP 1 Literature Review on Near-Field UHF RFID anti-collision algorithms. • 3D printable modular casing model developed in Blender/Fusion 360. • Complete Figma mobile UI prototype and BLE protocol specification.
Phase 2: Development (Months 4–6)	• Hardware Fabrication & Wiring • Embedded Firmware Development • Mobile Application Implementation	• Functional physical prototype with integrated UHF reader and zipper triggers. • ESP32-S3 FreeRTOS firmware running dual-interface updates (SPI E-Paper + BLE advertising). • Functional Flutter cross-platform mobile application for Android/iOS with SQLite/Firebase sync.
Phase 3: Post-Dev (Months 7–9)	• Real-World Bag Validation • Power Profiling & Edge Cases • Thesis Documentation & Competitions	• Empirical testing matrix across 30 distinct item materials (metals, liquids, plastics). • Battery consumption profiling report (Sleep vs Active). • Final FYP 2 Thesis and manuscript prepared for Innovate Malaysia / SCOPUS conference.

- **Embedded Communication Middleware:** Bluetooth Low Energy (BLE 5.0) Generic Attribute Profile (GATT) custom services for instant local telemetry pairing, augmented with lightweight MQTT/HTTP for cloud backup.
- **KiCad EDA:** Utilized for schematic drafting and custom power breakout PCB routing to ensure clean signal integrity between the RF reader and main MCU.

5 Hardware Architecture and Bill of Materials (BOM)

To satisfy the strict budget ceiling of RM 1,000, BagSense leverages cost-effective micro-UHF transceiver chips paired with passive ultra-low-cost sticker tags. The system operates on a smart event-driven power profile powered by a lightweight rechargeable lithium polymer cell.
Budget Note: At approximately **RM 571.00**, the project comfortably sits well under the RM 1,000 budget cap while leaving sufficient financial buffer for spare passive tags and mechanical prototyping iterations.

6 Target Performance Benchmarks

- **Inventory Detection Recall Rate:** $\geq 98\%$ recognition of all tagged items inside the closed backpack across varying tag orientations.
- **Interrogation Latency:** Complete multi-tag inventory scan and manifestation update within ≤ 1.8 seconds from zipper engagement.

Component Description	Technical Specification	Qty	Cost (RM)
Main Microcontroller Unit	ESP32-S3 (Dual-Core 240 MHz, BLE 5.0 + Wi-Fi, 8MB Flash)	1	35.00
Micro UHF RFID Reader Module	YRM100 / MagicRF M100 Nano UHF Module (865–868 MHz)	1	240.00
Near-Field Internal Antenna	Compact Ceramic / Flexible PCB UHF Patch Antenna (50 Ω)	1	45.00
Glanceable External Display	2.9-inch 3-Color or Monochrome E-Paper Display (SPI)	1	58.00
Passive UHF RFID Transponders	Impinj Monza / Alien Higgs-3 UHF Adhesive Tags	50	28.00
Bag State Sensing	Miniature Hall-Effect Sensor + Neodymium Magnet (Zipper)	2	8.00
Secondary Cavity Sensor	Micro Time-of-Flight (VL53L0X) or Force-Sensitive Resistor	1	14.00
Power Management & Battery	3.7V 3000 mAh Li-Po Cell + TP4056 Step-Up Protected BMS	1	38.00
Chassis & Cavity Materials	Shielding Fabric Mesh (Faraday attenuation) + 3D Filaments	1 Lot	45.00
Harness, Connectors & Switches	JST-PH Plugs, Silicone Wiring, Tactile Power Button	1 Lot	20.00
Test Base Backpack	Multi-compartment standard student backpack for retrofit	1	40.00
Total Projected Cost			**RM 571.00**

- **RF Containment / Zero-Leakage Radius:** Stray reading suppression ensures items placed farther than 25 cm outside the exterior of the bag are excluded from false-positive detection.
- **Energy Autonomy:** Minimum 5 days of normal daily use (assuming 25 zipper opening/closing scan cycles per day) utilizing the bistable zero-power static display properties of the E-Paper panel.

7 Conclusion and Academic Value for KICT

Project BagSense transforms an everyday object into an active, intelligent computing platform. By combining non-line-of-sight UHF RFID physics, ultra-low-power edge computing on the ESP32-S3, and cross-platform human-computer interaction (via external E-Paper and mobile BLE), this project addresses a tangible real-world problem with an elegant, scalable, and patentable architecture.

Under your supervision, this project is uniquely positioned to achieve top academic honors:

- **High Competition Readiness:** The concept is primed for nationwide innovation showcases such as *Innovate Malaysia Excellence Award*, *CITREX*, and the *IIUM Kulliyyah Exhibition (IRIIE)*.
- **Publication Potential:** The RF cavity attenuation models and multi-tag anti-collision edge algorithms yield original empirical data suitable for indexing in IEEE or SCOPUS-indexed Internet of Things conferences.
- **Direct Relevance to KICT:** It demonstrates the convergence of IoT embedded firmware, wireless sensor networks, and human-computer interface design.

I welcome the opportunity to discuss this proposal further and refine the technical milestones in accordance with your laboratory’s research direction.